

2nd order ODEs

$$a(x) y'' + b(x) y' + c(x) = d(x)$$

standard form

$$y'' + p(x) y' + q(x) y = r(x) \quad (*)$$

homogeneous if $r(x) = 0$

Homogeneous case

superposition principle

$y_1(x)$ and $y_2(x)$ are solutions of $(*)$

$\Rightarrow$ so is $\alpha \cdot y_1(x) + \beta \cdot y_2(x)$

example: $y'' + y = 0$ $\sin(x)$ and $\cos(x)$ solutions

$\Rightarrow 0,5 \cos(x) + 0,1 \cdot \sin(x)$ is also a solution

Statement: if y_1 and y_2 are independent (linearly) $\Rightarrow \alpha \cdot y_1 + \beta \cdot y_2$ is the general solution!

• $\sin(x)$ and $\cos(x)$ are independent

• $\sin(x)$ and $2 \cdot \sin(x)$ are not...

Def: linear independence:

$$\tilde{y} = k_1 \cdot y_1(x) + k_2 \cdot y_2(x) = 0 \quad \forall x \in I$$

$$\Rightarrow k_1 = k_2 = 0$$

• if $y_1 = \sin(x)$ $y_2 = 2 \cdot \sin(x)$

$$\Rightarrow \tilde{y} = (k_1 + 2k_2) \cdot \sin(x)$$

can be satisfied by $k_1 = 1, k_2 = -\frac{1}{2}$

• not true for non homogeneous DE

• not true for non-linear DE

Initial value problem

$$y'' + p(x) \cdot y' + q(x) = 0$$

find solution with $y(0) = k_1$ and $y'(0) = k_2$

example:

$$y'' + y = 0$$

find solution with $y(0) = y'(0) = 1$

solution:

$y_1 = \sin(x)$ and $y_2 = \cos(x)$ are

indep. solutions $\Rightarrow c_1 \cdot \sin(x) + c_2 \cos(x) = h(x)$
gen. solution

$$h(0) = c_2 = 1 \quad \Rightarrow \quad h(x) = \cos(x) + \sin(x)$$

$$h'(0) = c_1 = 1$$

